Math 5601 - Independent Study Proposal

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Proposed Topic: *Adaptive Quadrature Methods for Numerical Integration*

1. Goal of the Study

The goal of this independent study is to understand, implement, and analyze **adaptive quadrature algorithms** for numerical integration. Specifically, the objectives are to:

- Explain why adaptive quadrature is more efficient than fixed-step composite rules for functions with nonuniform smoothness.
- Implement an adaptive quadrature routine (e.g., based on Simpson's rule or the 8-panel Newton–Cotes rule) in MATLAB.
- Compare its performance (accuracy and efficiency) against non-adaptive composite rules such as the trapezoidal rule and Simpson's rule.
- Explore how the error estimation and interval subdivision criteria affect convergence and computational cost.

2. Study Plan and Steps

1. Literature Review

Review textbook sections and papers discussing adaptive quadrature, especially the **quanc8** routine described by Forsythe, Malcolm, and Moler. Study error estimation techniques in Newton–Cotes and Gaussian quadrature formulas, and understand the relationship between interval refinement and error control.

2. Algorithm Analysis

Outline the general adaptive quadrature framework, including branching, bounding, and stopping criteria. Derive and explain how the local error estimate E_c is computed by comparing single-interval and composite estimates. Examine the connection between adaptive quadrature and branch-and-bound methods.

3. Implementation

Write a MATLAB or Python implementation of an adaptive Simpson's rule or the 8-panel Newton–Cotes rule. Include adjustable parameters for error tolerance ε and minimum subinterval size δ.

4. Testing and Evaluation

Test the implementation on several functions, including:

- Smooth functions, e.g. $f(x) = \sin(x)$
- Rapidly varying functions, e.g. $f(x) = e^{-x^2}$
- Functions with localized nonsmoothness, e.g. $f(x) = \sqrt{x^2 + \epsilon}$ for small ϵ

Compare the adaptive routine's performance (total function evaluations and achieved accuracy) with composite Simpson's rule.

5. Analysis and Discussion

Analyze how the adaptive method automatically allocates smaller subintervals where the function varies rapidly. Discuss limitations, such as cases where the error tolerance cannot be met due to rounding or minimum interval constraints.

6. Summary and Conclusions

Summarize the advantages of adaptive quadrature in practical computational settings. Reflect on the balance between accuracy and efficiency in numerical integration routines.

3. References

1. Forsythe, G. E., Malcolm, M. A., & Moler, C. B. *Computer Methods for Mathematical Computations.* Prentice-Hall, 1977.
2. Burden, R. L., & Faires, J. D. *Numerical Analysis.*
3. Sauer, T. *Numerical Analysis.* 3rd Edition, Pearson, 2018.
4. Press, W. H. et al. *Numerical Recipes in C: The Art of Scientific Computing.* Cambridge University Press, 1992.
5. NETLIB Repository: `quanc8.f` – `http://www.netlib.org/fmm/quanc8.f`